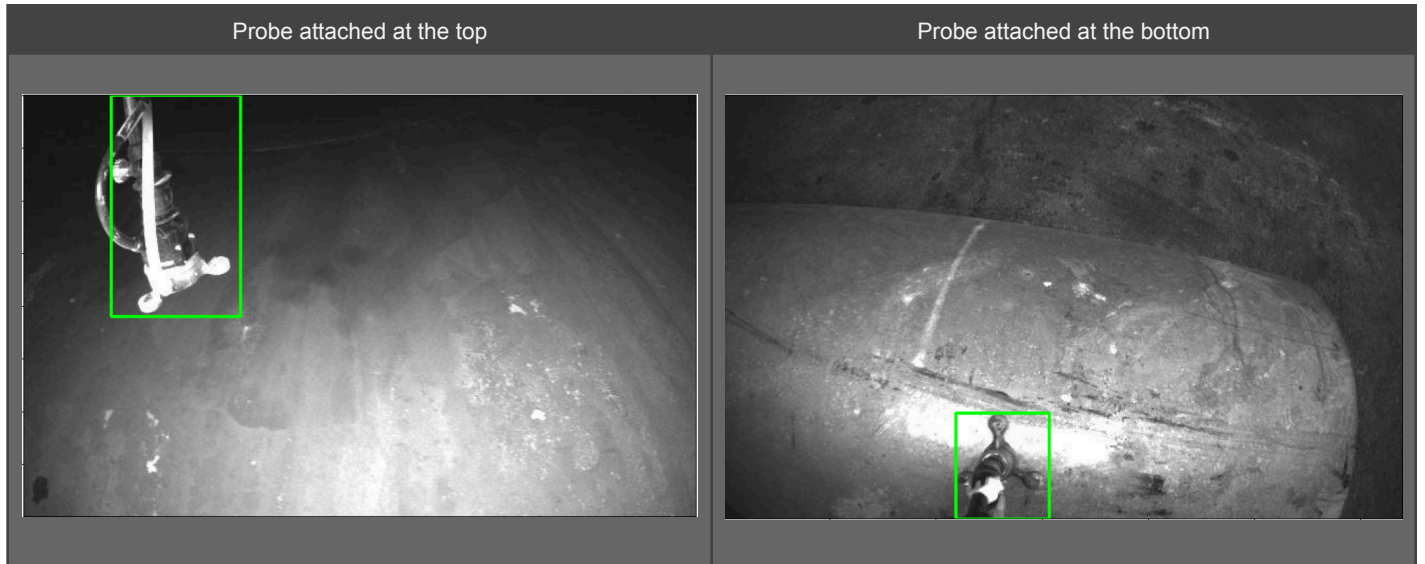


Candidate Assignment: Probe Detection with Deep Learning

Thank you for your interest in our position! To help us assess your skills and approach to solving a practical problem please complete the following assignment. The test is designed to evaluate your understanding of deep learning, system adaptation, and data processing.

To this end we prepared an annotated set of images taken by the Elios3 drone. Each image contains an ultrasonic thickness measurement device used to measure the thickness of metallic surfaces. Here is an example of such images with a bounding box encompassing the probe.



The goal of this project is to build a deep learning based system that will take an image as input and will output the bounding box of the probe if a probe has been detected. If no probe has been detected, the system must notify it. The detail of the output format is left at your discretion.

This assignment will help us evaluate your ability to:

- Use an existing deep learning-based system to detect probes in images.
- Preprocess and utilize provided data.
- Train or fine-tune a system for optimal performance.
- Evaluate the results and provide actionable insights.

Dataset

You are provided with the following:

1. A **probe_images** folder containing images.
2. A **probe_labels.json** JSON file containing the annotations for the dataset. The “**annotations**” array of the JSON specifies the regions in the images that correspond to probes (bounding boxes). Each annotation is linked to an image with the “**image_id**” field. The “**images**” array in the JSON specifies the file path for each image within the **probe_images** folder.

[You can download the dataset zip file here](#) containing the images and the JSON file.

Assignment Tasks

1. **System Selection and Setup**
 - Select a relevant existing deep learning system for the task.
 - Justify your choice of system in a short paragraph.
2. **Training and Fine-Tuning**
 - Train or fine-tune the system on the provided dataset.
 - Document the training process, including any hyperparameters used (e.g., learning rate, batch size, ...).
3. **Evaluation**
 - Evaluate the performance of the system on the dataset using appropriate metrics (e.g., precision, recall, F1-score, IoU, ...).
 - Evaluate the inference runtime of the system.
 - Provide a short analysis of the results: Highlight strengths, weaknesses, and any observable trends.
 - Please create a python script named “inference.py” (raw python, no Jupyter notebook) that takes as parameter the path of a folder containing image files. The script should iteratively go through each image **one by one**, detect the probe bounding box if any, draw the bounding box on the image and either save the resulting image to an output folder or show the resulting image in a window.
4. **Future Improvements**
 - Suggest at least two ways to improve the system’s performance or robustness in practical applications.
 - Suggest inference runtime improvements that can be done to make it run on an IoT board like the NVIDIA Jetson board family.

Deliverables

By the deadline, please submit the following:

1. A brief report (ex: PDF format) summarizing your approach, system selection, results, and insights.

2. The code used to complete the assignment (organized and commented, in a GitHub repository or as a zip file). Please don't forget to include the **inference.py** script discussed in the assignment tasks section above.
3. The trained weights of the network together with the detailed instructions for us to run the inference on our validation set.
4. Optional: Any visualizations or graphs that support your analysis.